

## EXPERIMENT 13 – Geographic Data Analysis

### Question

Analyze the relationship between population and average temperature for different cities.

### Dataset

| City   | Population | Avg. Temperature | Elevation |
|--------|------------|------------------|-----------|
| City A | 500000     | 75               | 1000      |
| City B | 700000     | 68               | 800       |
| City C | 600000     | 80               | 1200      |

The supplied data contains city names, population, temperature, and elevation, but does not contain latitude/longitude coordinates. Therefore, an actual geographic map cannot be generated from the given dataset.

### R Code

```
# Experiment 13 - Geographic Data

city <- c("City A", "City B", "City C")

population <- c(500000, 700000, 600000)

temperature <- c(75, 68, 80)

elevation <- c(1000, 800, 1200)

data <- data.frame(city, population, temperature, elevation)

# Question 1 - City Visualization

barplot(population,
        names.arg = city,
        main = "Population of Cities",
        xlab = "City",
        ylab = "Population",
        col = "skyblue")
```

```
# Question 2 - Scatter Plot
```

```
plot(temperature, population,  
     pch = 19,  
     main = "Temperature vs Population",  
     xlab = "Average Temperature",  
     ylab = "Population")
```

```
text(temperature, population,  
     labels = city,  
     pos = 3)
```

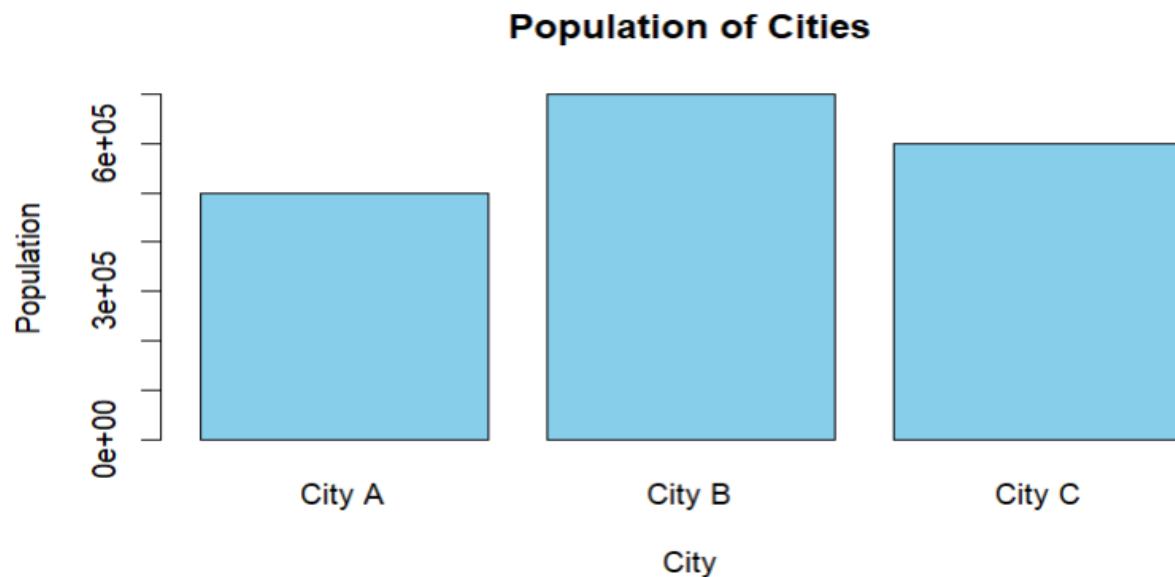
```
# Question 3 - Table
```

```
print(data)
```

### Question 1

Create a map-style visualization to show the distribution of cities.

### Output 1

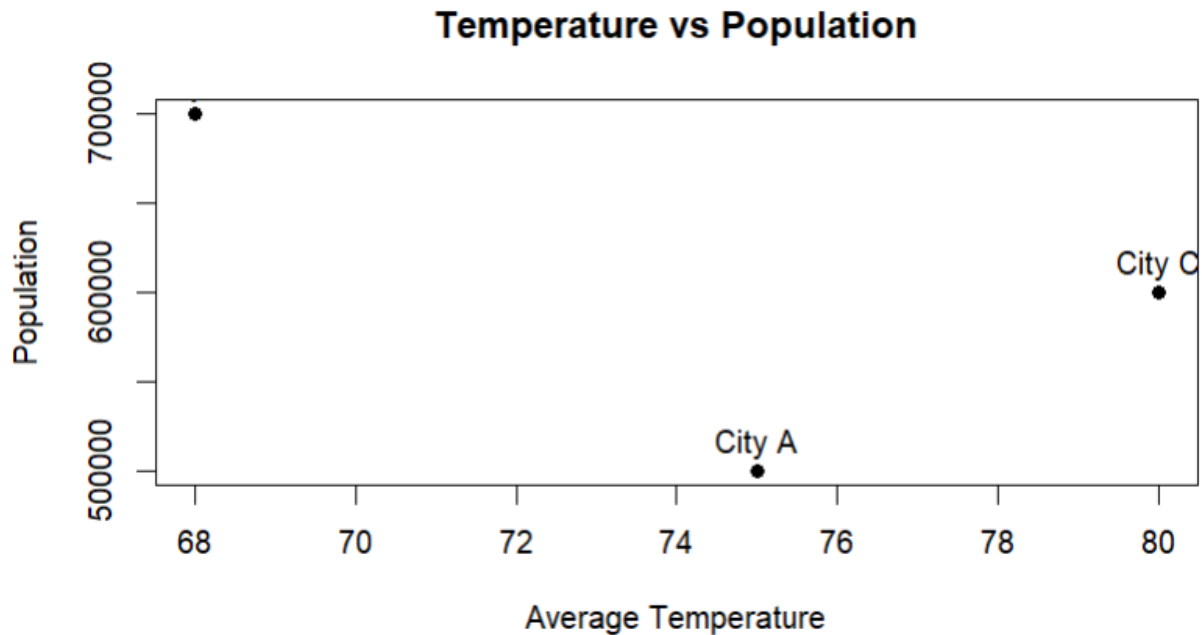


A city population visualization is generated because the supplied dataset does not contain geographic coordinates required for an actual map.

## Question 2

Generate a scatter plot to explore the relationship between average temperature and population.

### Output 2



The scatter plot shows that City B has the highest population, while City C has the highest average temperature.

## Question 3

Display the geographic data in a table.

### Output 3

```
> # Question 3 - Table
> print(data)
  city population temperature elevation
1 City A      5e+05         75        1000
2 City B      7e+05         68         800
3 City C      6e+05         80        1200
```